

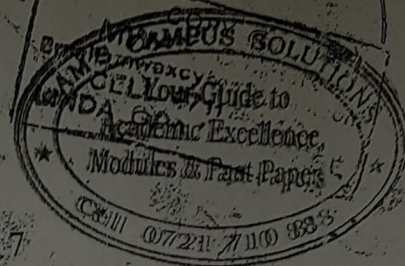
ECU 104



KENYA UNIVERSITY

UNIVERSITY EXAMINATIONS 2016/2017

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (ENGINEERING)



ECU 104: MATHEMATICS FOR ENGINEERS I

DATE: Wednesday 7th December 2016

TIME: 11.00a.m-1.00p.m

INSTRUCTIONS:

Answer question ONE and any other TWO

Question One (30 marks)

- Find the equation of the straight line, which passes through the point of intersection of the line $3x-2y=7$ and $5x+y=3$ and is perpendicular to the line $4x-5y=8$. (5 marks)
- Show that the line $x+3y=15$ is a tangent to the circle $x^2+y^2-8x+6y=15$ and find the point of contact. (5 marks)
- State the vertex, focus, axis and directrix of the parabola $x^2-6x-8y=31$ and hence sketch it. (5 marks)
- Determine the Cartesian equation for the curve with polar equation $r + 2r\cos\theta\sin\theta = 1/r$. (5 marks)
- Find the equation of the plane through the points $(1,1,1)$, $(1,3,2)$ and $(2,3,1)$. (5 marks)
- A particle moves in a straight line such that after t seconds its displacement S metres from a fixed point is given by $s = 4t - 2t^2 - 5t^3$. Find
 - Displacement at $t=2$ (1 mark)
 - Velocity at $t=3$ (2 marks)
 - The acceleration at $t=2$ (2 marks)

Question Two (20 marks)

$y=mx+c$

- Determine the locus of all points $P(x,y)$ that are equidistant from the point $(4,6)$ and the line $y=5$. (5 marks)

$mp-q+c$
 $\sqrt{m^2+1}$

$y=5$

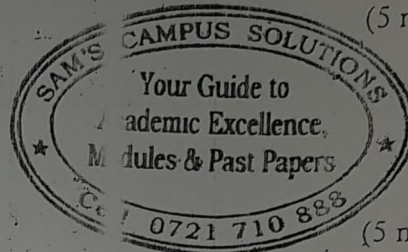
parabola

(5 marks)

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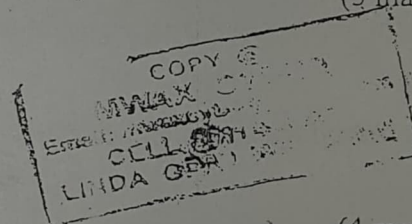
- b) Find the shortest distance from the point $(25,5)$ to the circle $x^2 + y^2 - 2x + 4y = 20$ (5 marks)
- c) The centre of a circle is on the line $y+5x=13$. The circle passes through the points $(-2, 14)$ and $(15,3)$ with centre on the line. Determine its equation in standard form. (5 marks)
- d) The vertex of a parabola is $V(3,-2)$ and its directrix is the line $y=-5$. Find the equation of the parabola and determine its focus. (5 marks)

Question Three (20 marks)



- a) Describe the following conic sections
- i) $4x^2 - 9y^2 + 40x + 36y + 100 = 0$ (5 marks)
- ii) $16x^2 + 36y^2 - 128x + 72y - 284 = 0$ (5 marks)
- b) Determine the polar equation of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ in the form $r + a\cos\theta + b\sin\theta = c/r$ (5 marks)
- c) Find the Cartesian equation for the curve with polar equation $r^2 + 2r^2\cos\theta\sin\theta = 1$ (5 marks)

Question Four



- a) Let $\underline{u}=(2, -7, 1)$, $\underline{v}=(-3,0,4)$, $\underline{w}=(0,5,-8)$. Find $2\underline{u} + 3\underline{v} - 5\underline{w}$ (4 marks)
- b) Determine k so that the vectors $\underline{u}$ and $\underline{v}$ are orthogonal where $\underline{u}=(1, k, -3)$ and $\underline{v}=(2, -5, 4)$ (5 marks)
- c) Find parametric and symmetric equations of a line passing through the points $P=(5,4,-3)$ and $Q=(1, -3, 2)$. (7 marks)
- d) Determine x , y , and z if $(-1, 3, 3) = x(1, 1, 0) + y(0, 0, -1) + z(0, 1, 1)$ (4 marks)

Question Five

- a) Find the angle between the following two planes, $2x+y-z=1$ and $x-2y-2z=2$. (5 marks)
- b) A river 4 km wide flows south with a current of 5km/h. What speed and heading should a motorboat assume to travel directly across the river from east to west in 20min? (5 marks)

ECU 104



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

ECU 104: MATHEMATICS FOR ENGINEERS 1

DATE: THURSDAY 13TH DECEMBER 2018

TIME: 2.00 P.M. – 4.00 P.M.

INSTRUCTIONS:

Answer Question ONE and any other TWO

Question One (30 marks)

1. (a) Calculate the shortest distance from the point (1, 2, 1) to the plane $x+2y+3z=2$ (5 marks)

(b) Determine the co-ordinate where the line $\frac{x-3}{2} = \frac{1-y}{1} = \frac{z}{3}$ meets the plane $2x+y+z=9$ (5 marks)

(c) Calculate the length of the tangent to the circle whose equation is $x^2+y^2=4x+8y+13$ from point (7, 9). (5 marks)

(d) Find the equation of a line Passing through the Point of Intersection of the line $2x+7y=3$ and $3x-2y=-8$ and is perpendicular to the line $5x-y+8=0$ (5 marks)

(e) Convert the Polar equation

$r=4 \sin \theta$ into cartesian form and hence describe it. (5 marks)

(f) Describe the conic Section fully $x^2-y^2-4x+2y+11=0$

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Question Two (20 marks)

2. (a) Determine the Parametrics, Symmetric and vector equation of a line passing through the points (2, 3,-4) and 1, -3,4). (5 marks)

(b) Find the equation of the plane passing through the following points (2, 0, 2), (1, 1, 1) and (-1, 1, 2) (5 marks)

- (c) Calculate the angle between the planes $2x+y-2z=1$ and $x-2y-2z=2$. (5 marks)
- (d) Verify that $(\underline{U} \times \underline{V}) \cdot \underline{W} = \underline{U} \cdot (\underline{V} \times \underline{W})$ if $\underline{U}$, $\underline{V}$ and $\underline{W}$ are vectors in 3 dimensions (5 marks)

Question Three (20 marks)

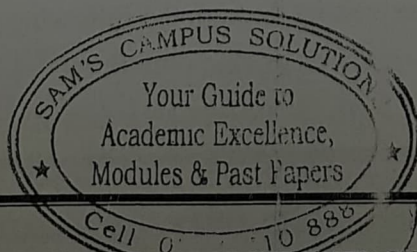
- (a) Given that one focus of a hyperbola is at $(1, -3)$ and the corresponding directrix is the line $y=2$, determine the equation of the hyperbola if its eccentricity is $3/2$ (7 marks)
- (b) If a hyperbola passes through the points $(12, 9)$ and has foci at the point $(0, 4)$ and the origin state its equation (7 marks)
- (c) Given that the general equation of a circle is $x^2 + y^2 + Dx + Ey + c = 0$ show that the centre of the circle is $\left(-\frac{D}{2}, -\frac{E}{2}\right)$ and the radius is given by $r = \sqrt{\frac{D^2}{4} + \frac{E^2}{4} - c}$ (6 marks)

Question Four (20 marks)

- (a) Define an ellipse and eccentricity of an ellipse (5 marks)
- (b) Discuss and sketch the graph of the equation $4x^2 + 9y^2 - 32x - 36y + 64 = 0$ (7 marks)
- (c) The end points of the major and minor axes of an ellipse are $(-1, 1)$, $(5, 1)$, $(2, 3)$ and $(2, -1)$. Give the equation of the ellipse in standard form and find its foci, eccentricity and directrices. (8 marks)

Question Five (20 marks)

- (a) (i) Define a parabola (2 marks)
- (ii) discuss the curve $2x^2 + 5y - 3x + 4 = 0$ (7 marks)
- (b) Find the equation of the curve such that the distance from any point $P(x, y)$ on the curve to the line $x=3$ is the same distance to the point $(4, 0)$. Hence sketch it (4 marks)
- (c) Consider the lines $7x+2y=7$ and $2x-3y=27$, find
- (i) the angle between the lines (3 marks)
- (ii) the distance from their point of intersection to the line $x=3y+5$ (4 marks)





KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (ENGINEERING)

ECU 104: ENGINEERING MATHEMATICS 1

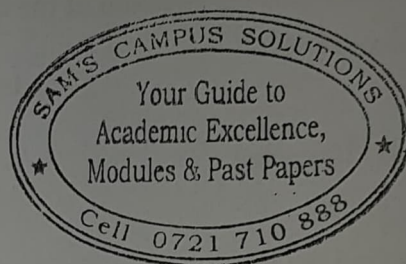
DATE: Wednesday, 21st February 2018

TIME: 8.00a.m-10.00a.m

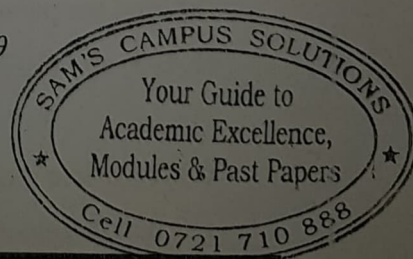
INSTRUCTIONS:

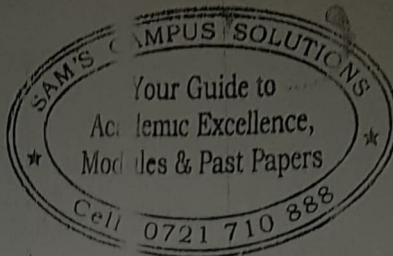
Answer Question ONE and any other TWO

Question 1 (30marks)



1. (a) Find the equation of the line that passes through the point of intersection of the lines $3x-2y+8=0$ and $2x+7y=3$ and is perpendicular to the line $5x+y-8=0$. (5marks)
- (b) Determine the equations of the two parabolas with a common vertex (2,4) and Passes through the origin. (5marks)
- (c) A plane passes through the origin, points (1,2,3) and (3,-2,1). Determine the equation of that plane (5marks)
- (d) Calculate the angle between the planes whose equation are $3x-y+2z=7$ and $x+4y-3z=6$ (5marks)
- (e) Identify the conic section represented by the equation $5x^2-4y^2+20x+8y=4$ (5marks)
- (f) Convert into a cartesian equation the polar equation $r = 4 \sin \theta$ (5marks)





Question 2 (20marks)

2. (a) Find the equation of the curve such that the distance from any point $p(x,y)$ on the curve to the line $X=3$ is the same distance to the point $(4,0)$. (7marks)
- (b) Identify the Vertex, focus axis of symmetry and equation of the directrix of the curve whose equation is $2x^2 + 5y - 3x + 4 = 0$ (7marks)
- (c) Determine the equation of the circle whose centre is $(-1,1)$ and it is tangent to the line $x+2y=4$. (6marks)

Question 3 (20marks)

3. (a) Find the general equation of the line that has X-intercept -3 and passes through the centre of the circle with equation $x^2 + y^2 - 4x + 10y + 26 = 0$ (8marks)
- (b) The line $x-2y=0$ cuts the circle $x^2 + y^2 - 8x + 6y - 15 = 0$ at the points P and Q state
- (i) The co-ordinates of P and Q (4marks)
 - (ii) The equation of the circle through points P and Q and point $(1,1)$ (4marks)
 - (iii) The equation of the tangent to the circle in b (ii) above at the point $(1,1)$ (4marks)

Question 4(20marks)

4. (a) Determine the values of x,y and z if $(2,-3,4) = x(1,1,1) + y(1,1,0) + z(1,0,0)$ (5marks)
- (b) Find the parametric and Symmetric equations of the line passing through the points $(5,4,-3)$ and $(1,-3,2)$. (6marks)
- (c) A particle moves along the space curve $r = (t^2 + t)i + (3t - 2)j + (2t^3 - 4t^2)k$. Find at time $t=2$, its (i) Velocity (3marks)

(ii) Acceleration

(3marks)

(iii) Magnitude of acceleration

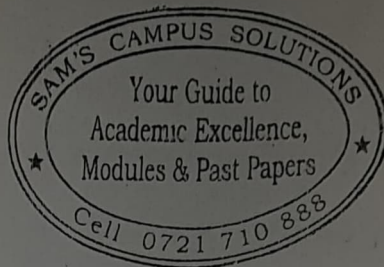
(3marks)

Question 5 (20marks)

Type equation here.

(a) Discuss the conic section

$$4x^2 + 9y^2 - 32x - 36y + 64 = 0$$



(8marks)

(b) State the centre, vertices, foci and asymptotes of the hyperbola

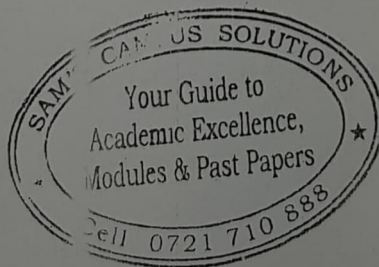
$$x^2 - y^2 - 4x + 2y + 11 = 0$$

(8marks)

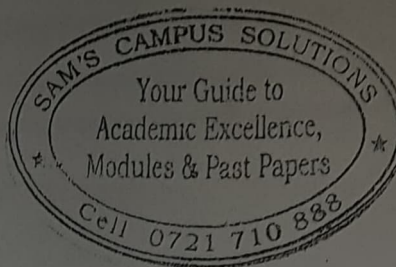
(c) Let $\underline{a} = (2, 1, -3)$, $\underline{b} = (1, -2, 1)$ and $\underline{c} = (-1, 1, -4)$.

Find $\underline{a} \cdot (\underline{b} \times \underline{c})$

(4marks)



Answer ALL questions



QUESTION 1

- (a) Let $u = (2, -7, 1)$, $v = (-3, 0, 4)$, $w = (0, 5, -8)$. Find $2u + 3v - 5w$
- (b) Find x, y and z if $(2, -3, 4) = x(1, 1, 1) + y(1, 1, 0) + z(1, 0, 0)$
- (c) Write the line $\frac{x-2}{2} = \frac{y-3}{3} = \frac{z-4}{5}$ in the form $v = a + \lambda u$

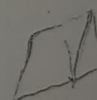
QUESTION 2

A particle moves in a straight line such that after t seconds its displacement s metres from a fixed point, is given by $s = 4t - 2t^2 - 5t^3$

Find

- (i) The velocity at $t = 3$
- (ii) When the velocity is zero
- (iii) Acceleration at $t = 2$

Gradient $\frac{y_2 - y_1}{x_2 - x_1}$



QUESTION 3

Find the equation of the plane through the points $(1, 0, 1)$, $(2, 1, 3)$, $(-4, 1, -1)$

QUESTION 4

Find the angle between the vectors $2i - j + k$ and $i + j - k$

ECU 104: ENGINEERING MATHEMATICS I

CAT 1

Answer all questions

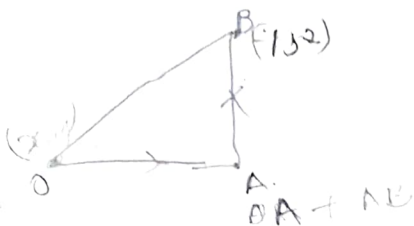
- Q1. (a) Find the centre and radius of the circle $x^2 + y^2 + 4x - 6y - 3 = 0$
- (b) Find the equation of the circle whose centre is on the x -axis and which passes through the points $(0, 3)$, $(4, 1)$
- Q2. Find the equation of the parabola with focus $(1, 1)$ and vertex $(-1, 1)$
- Q3. Find the centre, foci and vertices of the ellipse $\frac{(x+1)^2}{9} + \frac{(y-2)^2}{5} = 1$
- Q4. Find the centre, vertices, foci and asymptotes of the hyperbola

$$9x^2 - 16y^2 - 18x + 32y - 151 = 0$$

$$(y-k) = \pm \frac{b}{a}(x-h)$$

CAT 2

Answer ALL questions



1. (a) Find the initial point of the vector $-i + 2j$ if the terminal point is $(-1, -2)$
 (b) Write the vector $(1, 0)$ as a linear combination of vectors $(1, -3)$ and $(-1, 4)$
2. (a) Find the cosine of the angle between the vectors $v = (1, -1, 2)$ and $w = (2, 1, -1)$.
 Then find the projection of v on w . $\text{proj}_w v = |v| \cos \theta$
 (b) Find the parametric and the symmetric equations of the line passing through the points $(-1, 3, 7)$ and $(4, 2, -1)$.
3. (a) Find an equation of the plane through the point $(3, -7, 2)$ and perpendicular to the vector $(2, 1, -3)$.
 (b) Find the distance between the planes $x + y + 2z = 2$ and $x + y + 2z = 4$.
4. A stone is thrown vertically upwards and its height S metres above the ground after t seconds is given by $S = 40t - 10t^2$. Find
 - (i) Its height above the ground after 2 secs.
 - (ii) Its velocity after 1 sec.
 - (iii) The instant at which the stone is 30m high.



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2020/2021

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING, CIVIL ENGINEERING AND ELECTRICAL ENGINEERING

ECU 104: ENGINEERING MATHEMATICS I

DATE: Tuesday, 9th March 2021

TIME: 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS:

ANSWER QUESTION ONE AND ANY OTHER TWO QUESTIONS

QUESTION 1 (COMPULSORY - 30 MARKS)

- a) (i) Find the equation of line perpendicular to the line $x - 4y = 7$ and passing through the point $(3, -4)$. (3 Marks)
- (ii) Find the centre and radius of a circle with equation $x^2 + y^2 - 8x + 2y + 8 = 0$. (3 Marks)
- b) (i) Find an equation of a parabola with vertex $(1, 2)$ and focus $(4, 2)$. (3 Marks)
- (ii) Find the Cartesian equation of the curve $r \sin^2 \theta = 4 \cos \theta$ and show that the curve is a parabola. What is the vertex, focus and the equation of the directrix of this parabola? (5 Marks)
- c) Show that the curve $16x^2 + 9y^2 - 64x - 54y + 1 = 0$ is an ellipse and find its centre and points of its major axis, end points of its minor axis and its foci. (6 Marks)
- d) Show that the curve $x^2 - y^2 - 4x + 8y - 21 = 0$ is a hyperbola and indicate its centre, foci, vertices and asymptotes. (4 Marks)
- e) Find the equation of the plane containing the points $(-1, 2, 1)$, $(0, -3, 2)$, $(1, 1, -4)$ (3 Marks)

- f) The motion of a point Q may be described by the equation $S = -t^3 + 10t^2 + 8t$, where S is distance in metres and t is time in seconds. Calculate. (2 Marks)
- The maximum velocity of the motion.
 - Acceleration of the motion after 3 seconds. (1 Mark)
 - The time at which velocity is zero. (1 Mark)

QUESTION 2 (20 MARKS)

- Find the equation of the line parallel to $3x + 2y = 5$ and passing through the point $(-1, 2)$. (3 Marks)
- Find the Cartesian equation of the curve $r = 2(\sin \theta + \cos \theta)$ and show that the curve is a circle. What is the centre and the radius of this circle? (7 Marks)
- Find polar coordinates of the point whose Cartesian coordinates are $(-2, 2\sqrt{3})$ (5 Marks)
- Identify the curve $x^2 + y^2 - 4x + 6y + 9 = 0$. (5 Marks)

QUESTION 3 (20 MARKS)

- Find the vertex, focus and the equation of the directrix of a parabola whose equation is $X^2 = 4x + 8y - 4$. (5 Marks)
- Determine the points at which the tangent at the point $(1, 2)$ to the circle $x^2 + y^2 = 5$ cuts the circle $x^2 + y^2 = 10$. (7 Marks)
- What type of conic is given by the equation $x^2 + 6x + 4y^2 - 8y = 3$. (3 Marks)
- Find the equation of the hyperbola with vertices $(0, \pm 8)$ and asymptotes $y = \pm \frac{1}{2}x$. (5 Marks)

QUESTION 4 (20 MARKS)

- Write $(1, -2, 5)$ as a linear combination of $(1, 1, 1)$, $(1, 2, 3)$, $(2, -1, 1)$. (7 Marks)
- Find the parametric equations and the symmetric equations of the line which passes through the point $(4, -1, 9)$ and is perpendicular to the plane $3x - 2y + z = 18$. (4 Marks)
- Find the angle between the vectors $\underline{v} = \underline{i} + \underline{j} + \underline{k}$ and $\underline{u} = \underline{i} - \underline{j} + \underline{k}$. (3 Marks)
- Find the distance from the point $(3, -8, 1)$ to the line $\frac{x-3}{3} = \frac{y+7}{-1} = \frac{z+2}{5}$. (6 Marks)

QUESTION 5 (20 MARKS)

- a) Two forces F_1 and F_2 act on the same body. It is known that F_1 has magnitude 3 Newtons and acts in the direction of $-i$, whereas F_2 has magnitude 2 Newtons and acts in the direction of the unit vector $\underline{u} = \frac{3}{5}\underline{i} - \frac{4}{5}\underline{j}$. What force F_3 must be applied to keep the body at rest? (4 Marks)
- b) A river 2.1km wide flows south with current of 3.1 km/hr. What speed and heading should a boat assume to travel across the river from east to west in 30 minutes? (4 Marks)
- c) Find the terminal point of the vector $5\underline{i} + 7\underline{j}$ if the initial point is $(-2, 3)$. (4 Marks)
- d) A particle moves in such a way that after t seconds its displacement S metres from a fixed point is given by $S = 4t - 2t^2 - 5t^3$

Find

- (i) The velocity at $t = 3$. (3 Marks)
- (ii) When the velocity is zero (3 Marks)
- (iii) The acceleration at $t = 2$. (2 Marks)